

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Java, JavaScript, TypeScript, C++, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing, authentication, rate limiting, audit logging

Systems & Data

Apache Spark / PySpark, Databricks, Spark UI profiling, distributed systems concepts, performance engineering, large-scale ETL

Databases & DevOps

MongoDB, PostgreSQL, SQLite, Redis, Docker, Git, GitHub Actions, CI/CD, Linux

AI-Assisted Dev & GenAI

Claude / Claude Code, GitHub Copilot, Cursor, ChatGPT / Codex, OpenAI / Groq APIs, RAG, LLM agents, PyTorch

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Engineered **multi-tenant NHS eTriage backend** for **8+ UK clinical clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade distributed service integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Owned backend reliability and quality across the eTriage stack: monitored production logs through feature releases and tenant integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and added **integration tests** for regression protection while remediating **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** on sensitive clinical data.
- Architected full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

Spark ETL Performance Optimization - NYC TLC Taxi Dataset

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized large-scale Spark ETL pipeline on the NYC TLC taxi dataset using **broadcast-join tuning, caching, partitioning, adaptive Spark settings, column pruning**, and precomputed fields; reduced shuffle volume and wall-clock runtime relative to the naive baseline.
- Used **Spark UI** stage-level metrics to localize shuffle and skew bottlenecks, then documented a repeatable optimization workflow comparing naive vs. tuned runs to make the methodology reusable across downstream analytics jobs.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation — using AI as a high-trust collaborator while keeping system behavior grounded in deterministic validation and a reviewer-facing trust UI.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, WebSockets, SQL

- Built cloud-native AI research assistant on Cloudflare Workers: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time — prototyped end-to-end with embedded SQL storage.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and individual tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023** — applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.